RAKSHAK

An on-device AI safety layer against cyberbullying, sextortion and child exploitation

Harm builds message by message. Protection should too.

RAKSHAK (Hindi: *protector*) is a proposed Android safety layer bound to a child's account and always visibly running. A small on-device classifier labels grooming and bullying cues in authorized messages; an encrypted per-contact memory tracks how those cues build over days; transparent, configurable rules decide when to nudge the child, alert a parent, or guide the parent through an official report.

TEAM DETAILS	
TEAM	Git Happens
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TRACK	Cybersecurity

01 · PROBLEM STATEMENT

Children are rarely harmed by one message. They are harmed by a pattern.

1 in 3
young people in 30 countries report being bullied online; 1 in 5 skipped school because of it
UNICEF poll, 2019 [1]

67,694
POCSO cases registered in India in 2023 — 38% of recorded crimes against children
NCRB 2023 [2]

~100/day
financial sextortion reports to the US hotline NCMEC in 2024
NCMEC, 2025 [3]

36
teenage boys known to have died by suicide after sextortion since 2021
NCMEC, 2025 [3]

Six days, one stranger — and how RAKSHAK's case score builds

ILLUSTRATIVE · FICTIONAL

DAY 1 · FLATTERY / AGE

> *you're so mature for your age. way cooler than anyone in your class*

WITHOUT RAKSHAK
No one is told

WITH RAKSHAK
LOGGED · RAPPORT
New contact, cue stored

+1.0 → SCORE 1.0

DAY 2 · SECRECY

> *don't tell your parents about our chats. they won't get it like I do*

WITHOUT RAKSHAK
No one is told

WITH RAKSHAK
LEVEL 1 · CHILD
Nudge: "Safe people don't ask you to hide chats."

0.9 + 2.0 → SCORE 2.9

DAY 4 · IMAGE REQUEST

> *send me a photo, just for me. then delete this chat*

WITHOUT RAKSHAK
No one is told

WITH RAKSHAK
LEVEL 2 · PARENT
Child warned, parent alerted

2.4 + 3.0×2 → SCORE 8.4

DAY 6 · THREAT / PAYMENT

> *pay ₹5,000 by tonight or your whole school sees what you sent*

WITHOUT RAKSHAK
Child faces it alone

WITH RAKSHAK
LEVEL 3 · URGENT
Parent guided to report and takedown

6.9 + 4.0 → SCORE 10.9

Scores use the planned default weights and 7-day half-life in §03 (×2 = secrecy then image request; Day 4 also meets the new-contact floor, Day 6 the threat floor). Intended design behaviour, not experimental results.

The same build-up drives bullying, where insults from one person or group repeat across days until a child withdraws. A frightened child often tells no one until it has escalated. Fixed word rules are brittle against this — the early messages contain no bad words, and spellings change — which is why RAKSHAK pairs a **learned cue classifier** with a **memory of each contact's history**.

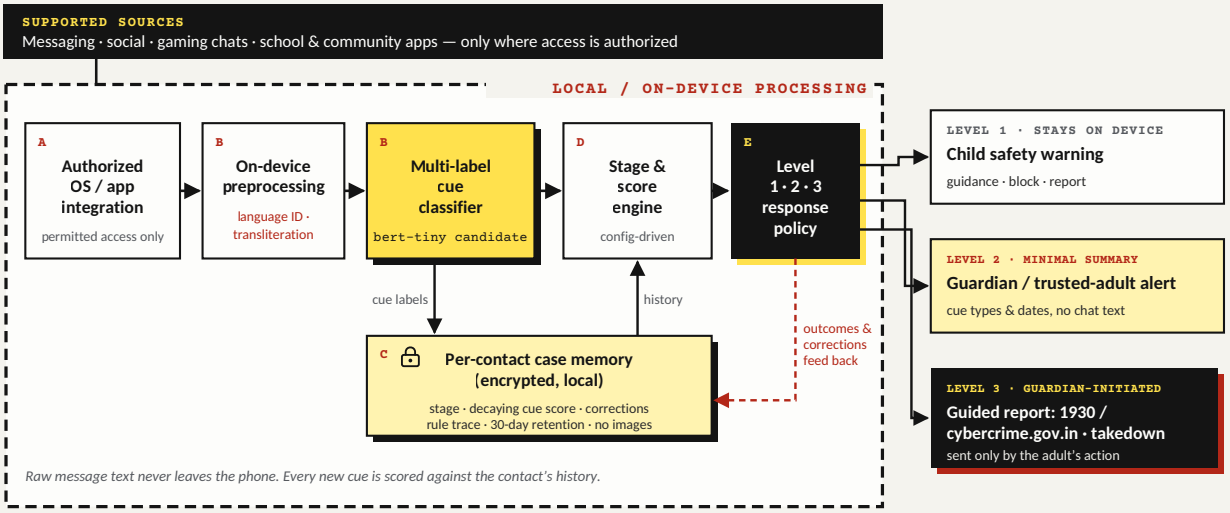
Children & teens
Primary users, most exposed to grooming, sextortion and bullying, and least likely to report early.

Parents
Alerted at Level 2–3 with a minimal summary, plus guided next steps.

Alternate trusted adult
Chosen at setup by child and parent together (e.g. other parent, relative, school counsellor). Alerts go here if the flagged contact is the parent, or if the child asks.

CITIES OF THE FUTURE For city digital-school systems, RAKSHAK can act as a privacy-first safety layer: school counsellors form the trusted-adult tier, and guided reports route to the city or state cyber cell.

A learned cue classifier, a per-contact memory, and transparent thresholds — all on the phone



Honest coverage. Android notification access [6] sees incoming notification previews only — likely not the child's own sent messages, not messages read while a chat is open, and not images. Our **first build step** is to measure this on a real phone across WhatsApp, Instagram and Snapchat and publish a coverage table. RAKSHAK is the **text-pattern layer**; it complements image-side protections such as Google Messages' on-device sensitive-content warnings [12], not replaces them.

CUE TAXONOMY · MULTI-LABEL CLASSIFIER OUTPUT

CUE	EXAMPLE (EN / TANGLISH)	W*
Flattery / age	*"so mature for your age"*	1
Secrecy	*"don't tell your parents" / "amma kitta sollaadha"*	2
Isolation	*"your friends don't get you like I do"*	2
Platform move	*"add me on snap / telegram"*	2
Image request	*"send a pic, just for me"*	3
Meeting request	*"come alone, don't tell anyone"*	3
Disable protection	*"delete that app / turn it off"*	4
Threat / payment	*"pay or everyone sees it"*	4
Insult / harassment	*"nobody wants you here"*	1

**Planned default weights, to be calibrated on validation data.*

CASE MEMORY & RISK ENGINE · SPECIFICATION

CONTACT	RAPPORT	SECRECY / ISOLATION	REQUEST	COERCION

Each contact has a case with a **stage** and a **score**. Each cue adds its weight; scores **decay** with a 7-day half-life. **Sequences multiply:** secrecy or isolation followed by an image or meeting request counts ×2.

Thresholds: ≥2 → Level 1 · ≥5 → Level 2 · ≥9 → Level 3.

Severity floors set a minimum level regardless of score: threat / payment → Level 3; image or meeting request from a contact first seen within 14 days → Level 2; disable-protection → Level 2.

Reductions: quotes or counterspeech ×0.3. A child's dismissal ×0.5, but never below Level 1 and never at Request or Coercion stage; dismissals are counted and shown in the Level 2 summary.

Bullying: a child-level harassment score sums insult cues across all senders; it routes to the parent and school counsellor, not a cybercrime report, unless threats or images are involved.

Possible links: an unknown contact messaging within 24 h of a block starts at the blocked case's stage, pending review.

Alert channel: Level 2/3 alerts go to a guardian app paired by QR at setup, as an end-to-end encrypted minimal payload.

All weights live in `one config file`; every alert stores its rule trace.

Labelling & training plan

- **~2,000 English + Tanglish messages** labelled with the cue taxonomy by **two annotators**, agreement measured (Cohen's κ), from PAN-2012 lines [9], public code-mixed text [10] and our scenarios.
- **Tanglish grooming examples are team-written fiction** — so the test set is blind scenarios written by outsiders.
- Toxicity corpora [5][8][10] support the **insult/threat classes only**.

Model choice

- **Baseline to beat:** character n-gram logistic regression.
- **On-device candidate:** bert-tiny [4]; an Indic model [11] tested for Tanglish. Final pick by **measured accuracy and latency**.
- **Stack:** PyTorch · Android (Kotlin) · LiteRT or ONNX Runtime Mobile · Keystore-encrypted DB.

WHY THIS MATTERS FOR POCSO Grooming rarely starts with an explicit message. RAKSHAK keeps flattery, secrecy and isolation cues in one case per contact, so a parent learns of the pattern at the request stage and can use the official reporting route under the POCSO Act, 2012 [7]. People, not the system, decide whether an offence occurred.

04 · THREAT MODEL

Who RAKSHAK defends against, and how each defence is tested

Assets: the child's safety · the local incident database · the alert channel · the parent account.

ADVERSARY / EVASION	MITIGATION	PLANNED TEST
Patient groomer	Per-contact memory with slow decay keeps weeks-long build-up in one case	Multi-week fictional scenarios
Scripted sextortion ring, many accounts	Floors: threat / payment → Level 3 at once; new contact's image request → Level 2. Post-block contacts start at the linked stage, if earlier messages were captured	Same script replayed from rotating accounts
Group of peer bullies	Child-level harassment score sums insults across senders → parent + counsellor	Multi-sender pile-on scenarios
Abusive parent	Alternate trusted adult receives alerts if the flagged contact is the parent	Routing test with parent as sender
Physical access to the phone	Encrypted DB behind device lock; no images or full chats stored	Data-at-rest inspection
Child pressured to disable	Disable-protection cue floors at Level 2; revoking RAKSHAK's permission alerts the parent	Permission-revocation test
Coached dismissal	Dismissals cannot clear Request/Coercion cases or drop below Level 1; count shown to parent	Scripted "just dismiss it" scenarios
Spoofed or intercepted alert	QR-authenticated pairing; end-to-end encrypted minimal payload	Replay / tampering test on alert channel
Misspelling, leetspeak, emoji, Tamil script	Normalisation + transliteration before classification	Adversarial spelling / emoji set
App switching	Platform-move cue raises the case (if the earlier messages were captured)	Cross-app scenario
Voice notes & images	**Out of scope** — image-side tools such as Google Messages warnings [12]	Stated limitation

05 · INNOVATION

	PLATFORM MODERATION	BARK [13]	GOOGLE MESSAGES WARNINGS [12]	RAKSHAK
ANALYSES	Content on one platform	Texts, apps, email; alerts parents	Images (nudity)	**Text grooming & bullying cues**
PROCESSING	Platform servers	Not publicly stated	On device	**On device**
MEMORY	Platform-dependent	Not publicly stated	Per image	**Per-contact escalation stages**
LANGUAGES	Varies	Not publicly stated	Language-independent	**English + Indian code-mixed (planned)**
ALERTS	Remove, warn, ban	Parent alerts	Blur + warning to user	**Explainable, correctable, levelled**

06 · FEASIBILITY & IMPLEMENTATION PLAN

STEP 0 · FIRST
Coverage audit on a real phone: WhatsApp, Instagram, Snapchat → published table.

STAGE 1 · BEFORE → DURING
Before (if rules allow): labelling.
During 24 h: classifier vs baseline, case engine, simulation. No prep allowed → ~500 messages.

STAGE 2 · ANDROID
Native app: background analysis, parent pairing, guided reporting, battery tests.

STAGE 3 · PLATFORM
OS / school-system integration; Telugu, Hindi, Marathi & more.

07 · EVALUATION PLAN *— all planned, no results yet*

- **Three setups:** keyword-only · model per message · model + memory.
- **Blind scenarios** by outsiders who never saw the rules.
- **Hard negatives:** surprise-party secrets, friend banter, quoting the news.
- **Adversarial set:** misspellings, leetspeak, emoji, Tamil script.
- **Headline metric:** messages earlier than the keyword baseline, at a fixed false-alarm rate per child per week.

IMPACT, DATA & SAFEGUARDS

- **Impact:** a child warned at the secrecy stage, a parent alerted before the threat — while help still works.
- **Stored on device only:** cue labels, short redacted excerpts, timestamps, hashed contact ID; 30-day default retention; **images never stored**.
- **Level 3:** guided flow for the parent or alternate adult to cybercrime.gov.in / 1930 and takedown services [14], sent only by their action.
- Bound to a child account, always visible; designed for review under the DPDP Act 2023 & Rules [7].

Groomers, extortionists and bullies rely on one thing: that nobody connects the messages. RAKSHAK connects them — privately, on the child's own phone.

[1] UNICEF online bullying poll, 2019
[2] NCRB Crime in India 2023 (Down To Earth)
[3] NCMEC 2024 in Numbers
[4] bert-tiny (MIT)
[5] Civil Comments (CC0)
[6] Android NotificationListenerService
[7] POCSO Act 2012 · DPDP Act 2023
[8] Jigsaw Wikipedia toxicity (CC0)
[9] PAN 2012 Sexual Predator ID
[10] DravidianCodeMix (CC BY 4.0)
[11] IndicBERT, AI4Bharat (MIT)
[12] Google Messages sensitive content warnings
[13] Bark
[14] cybercrime.gov.in · NCMEC Take It Down